Experiment 05: Ridge and Lasso Regularization

Python program

```
# Experiment 05: Ridge and Lasso Regularization

# 1. Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score

# 2. Load Diabetes dataset
data = load_diabetes()

df = pd.DataFrame(data.data, columns=data.feature_names)
df['target'] = data.target

# 3. Display first five records
print("First Five Records:")
print(df.head())

# 4. Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# 5. Correlation Heatmap
plt.figure(figsize=(8, 5))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

# 6. Separate input and output
X = df.drop('target', axis=1)
y = df['target']

# 7. Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 8. Apply StandardScaler
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 9. Create models
models = {
    'OLS': LinearRegression(),
    'Ridge': Ridge(alpha=1.0),
    'Lasso': Lasso(alpha=0.1)
}

results = []
coefficients = []

# 10. Train and evaluate models
for name, model in models.items():

    model.fit(X_train, y_train)

    train_pred = model.predict(X_train)
    test_pred = model.predict(X_test)

    train_mse = mean_squared_error(y_train, train_pred)
    test_mse = mean_squared_error(y_test, test_pred)

    rmse = np.sqrt(test_mse)
    r2 = r2_score(y_test, test_pred)

    results.append([
        name, train_mse, test_mse, rmse, r2
```

```
    ])

    coefficients.append(model.coef_)

# 11. Display evaluation results
result = pd.DataFrame(results, columns=[
    'Model', 'Train MSE', 'Test MSE', 'Test RMSE', 'Test R2'
])

print("\nModel Comparison:")
print(result)

# 12. Compare coefficients
coef_df = pd.DataFrame(
    coefficients,
    columns=X.columns,
    index=['OLS', 'Ridge', 'Lasso']
)

print("\nCoefficient Comparison:")
print(coef_df)

# 13. Display coefficient graph
coef_df.T.plot(kind='bar', figsize=(10, 5))
plt.title("OLS vs Ridge vs Lasso Coefficients")
plt.xlabel("Features")
plt.ylabel("Coefficient Values")
plt.show()

# 14. Display completion
print("\nRidge and Lasso Completed Successfully!")
```